

Moore's Law Every year, you probably expect to pay at least a little more for most products and services. The opposite has been the case in the computer and communications fields, especially with regard to the hardware supporting these technologies. For many decades, hardware costs have fallen rapidly. Every year or two, the capacities of computers have approximately doubled inexpensively. This remarkable trend often is called Moore's Law, named for the person who identified it in the 1960s, Gordon Moore, co-founder of Intel—one of the leading manufacturers of the processors in today's computers and embedded systems. Moore's Law and related observations apply especially to • the amount of memory that computers have for programs, • the amount of secondary storage (such as solid-state drive storage) they have to hold programs and data over longer periods of time, and • their processor speeds—the speeds at which they execute their programs (that is, do their work). Similar growth has occurred in the communications field—costs have plummeted as enormous demand for communications bandwidth (that is, information-carrying capacity) has attracted intense competition. We know of no other fields in which technology improves so quickly and costs fall so rapidly. Such phenomenal improvement is truly fostering the Information Revolution.

Computer Organization Regardless of differences in physical appearance, computers can be envisioned as divided into various logical units or sections:

Input Unit This “receiving” section obtains information (data and computer programs) from input devices and places it at the disposal of the other units for processing. Most user input is entered into computers through keyboards, touch screens and mouse devices. Other forms of input include receiving voice commands, scanning images and barcodes, reading from secondary storage devices (such as hard drives, Blu-ray Disc™ drives and USB flash drives—also called “thumb drives” or “memory sticks”), receiving video from a webcam and having your computer receive information from the Internet (such as when you stream videos from YouTube® or download e-books from Amazon). Newer forms of input include position data from a GPS device, motion and orientation information from an accelerometer (a device that responds to up/down, left/right and forward/backward acceleration) in a smartphone or wireless game controller (such as those for Microsoft® Xbox®, Nintendo Switch™ and Sony® PlayStation®) and voice input from intelligent assistants like Apple Siri®, Amazon Echo® and Google Home®.

Output Unit This “shipping” section takes information the computer has processed and places it on various output devices to make it available for use outside the computer. Most information that's output from computers today is displayed on screens (including touch screens), printed on paper (“going green” discourages this), played as audio or video on smart phones, tablets, PCs and giant screens in sports stadiums, transmitted over the Internet or used to control other devices, such as self-driving cars, robots and “intelligent” appliances. Information is also commonly output to secondary storage devices, such as solid-state drives (SSDs), hard drives, DVD drives and USB flash drives. Popular recent forms of output are smartphone and game-controller vibration, virtual reality devices like Oculus Rift®, Sony® PlayStation® VR and Google Daydream View™ and Samsung Gear VR®, and mixed reality devices like Magic Leap® One and Microsoft HoloLens™.

Memory Unit This rapid-access, relatively low-capacity “warehouse” section retains information that has been entered through the input unit, making it immediately available for processing when needed. The memory unit also retains processed information until it can be placed on output devices by the output unit. Information in the memory unit is volatile—it's typically lost when the computer's power is turned off. The memory unit is often called either memory, primary memory or RAM (Random Access Memory). Main memories on desktop and notebook computers contain as much as 128 GB of RAM, though 8 to 16 GB is most common. GB stands for gigabytes; a gigabyte is approximately one billion bytes. A byte is eight bits. A bit is either a 0 or a 1.

Arithmetic and Logic Unit (ALU) This “manufacturing” section performs calculations, such as addition, subtraction, multiplication and division.

It also contains the decision mechanisms that allow the computer, for example, to compare two items from the memory unit to determine whether they're equal. In today's systems, the ALU is part of the next logical unit, the CPU. Central Processing Unit (CPU) This "administrative" section coordinates and supervises the operation of the other sections. The CPU tells the input unit when information should be read into the memory unit, tells the ALU when information from the memory unit should be used in calculations and tells the output unit when to send information from the memory unit to specific output devices. Most computers have multicore processors that implement multiple processors on a single integrated-circuit chip. Such processors can perform many operations simultaneously. A dual-core processor has two CPUs, a quad-core processor has four and an octa-core processor has eight. Intel has some processors with up to 72 cores. Today's desktop computers have processors that can execute billions of instructions per second. Secondary Storage Unit This is the long-term, high-capacity "warehousing" section. Programs or data not actively being used by the other units normally are placed on secondary storage devices (e.g., your hard drive) until they're again needed, possibly hours, days, months or even years later. Information on secondary storage devices is persistent—it's preserved even when the computer's power is turned off. Secondary storage information takes much longer to access than information in primary memory, but its cost per unit is much less. Examples of secondary storage devices include solid-state drives (SSDs), hard drives, read/write Blu-ray drives and USB flash drives. Many current drives hold terabytes (TB) of data—a terabyte is approximately one trillion bytes). Typical hard drives on desktop and notebook computers hold up to 4 TB, and some recent desktop-computer hard drives hold up to 15 TB.

Operating Systems. Operating systems are software systems that make using computers more convenient for users, application developers and system administrators. They provide services that allow each application to execute safely, efficiently and concurrently with other applications. The software that contains the core components of the operating system is called the kernel. Linux, Windows and macOS are popular desktop computer operating systems—you can use any of these with this book. The most popular mobile operating systems used in smartphones and tablets are Google's Android and Apple's iOS.

Windows—A Proprietary Operating System In the mid-1980s, Microsoft developed the Windows operating system, consisting of a graphical user interface built on top of DOS (Disk Operating System)—an enormously popular personal-computer operating system that users interacted with by typing commands. Windows 10 is Microsoft's latest operating system—it includes the Cortana personal assistant for voice interactions. Windows is a proprietary operating system—it's controlled by Microsoft exclusively. Windows is by far the world's most widely used desktop operating system.

Linux—An Open-Source Operating System The Linux operating system is among the greatest successes of the open-source movement. Open-source software departs from the proprietary software development style that dominated software's early years. With open-source development, individuals and companies contribute their efforts in developing, maintaining and evolving software in exchange for the right to use that software for their own purposes, typically at no charge. Open-source code is often scrutinized by a much larger audience than proprietary software, so errors often get removed faster. Open source also encourages innovation. There are many organizations in the open-source community. Some key ones are:

- Python Software Foundation (responsible for Python).
- GitHub (provides tools for managing open-source projects—it has millions of them under development).

Rapid improvements to computing and communications, decreasing costs and open-source software have made it much easier and more economical to create software-based businesses now than just a decade ago. A great example is Facebook, which was launched from a college dorm room and built with open-source software. The Linux kernel is the core of the most popular open-source, freely distributed, full-featured operating system. It's developed by a loosely organized team of volunteers and is popular in servers, personal computers and embedded systems (such as the computer systems at the heart of smartphones, smart TVs and automobile systems). Unlike that of proprietary operating systems like Microsoft's Windows and Apple's macOS, Linux source code (the program code) is available to the public for examination and modification and is free to download and install. As a result, Linux users benefit from a huge community of developers actively debugging and improving the kernel, and the ability to customize the operating system to meet specific needs. Apple's macOS and Apple's iOS for iPhone® and iPad® Devices Apple, founded in 1976 by Steve Jobs and Steve Wozniak, quickly became a leader in personal computing. In 1979, Jobs and several Apple employees visited Xerox PARC (Palo Alto Research Center) to learn about Xerox's desktop computer that featured a graphical user interface (GUI). That GUI served as the inspiration for the Apple Macintosh, launched in 1984. The Objective-C programming language, created by Stepstone in the early 1980s, added capabilities for object-oriented programming (OOP) to the C programming language. Steve Jobs left Apple in 1985 and founded NeXT Inc. In 1988, NeXT licensed Objective-C from Stepstone and developed an Objective-C compiler and libraries which were used as the platform for the NeXTSTEP operating system's user interface, and Interface Builder—used to construct graphical user interfaces. Jobs returned to Apple in 1996 when they bought NeXT. Apple's macOS operating system is a descendant of NeXTSTEP. Apple's proprietary operating system, iOS, is derived from macOS and is used in the iPhone, iPad, Apple Watch and Apple TV devices. In 2014, Apple introduced its new Swift programming language, which became open source in 2015. The iOS app-development community has largely shifted from Objective-C to Swift. Google's Android—the fastest growing mobile and smartphone operating system—is based on the Linux kernel and the Java programming language. Android is open source and free.

According to idc.com, as of 2018, Android had 86.8% of the global smartphone market share, compared to 13.2% for Apple.⁷ The Android operating system is used in numerous smartphones, e-reader devices, tablets, in-store touch-screen kiosks, cars, robots, multimedia players and more. Billions of Devices In use today are Billions of personal computers and an even larger number of mobile devices. The following table lists many computerized devices. The explosive growth of mobile phones, tablets and other devices is creating significant opportunities for programming mobile apps. There are now various tools that enable you to use Python for Android and iOS app development, including BeeWare, Kivy, PyMob, Pythonista and others. Many are cross-platform, meaning that you can use them to develop apps that will run portably on Android, iOS and other platforms (like the web).